


# Winning Space Race with Data Science

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https://github.com/nicholas-yap/spacex-data-science-capstone/

# Rubric Evidence Index

| Criterion | Evidence in this report |
|---|---|
| Q1 Repository and completed files | Public repository URL: pages 1, 40. |
| Q2 Final presentation in PDF format | Complete PDF: 40 pages. |
| Q3 Executive summary | Page 3: methods, results and limitations. |
| Q4 Introduction | Page 9: business question and scope. |
| Q5 Collection and wrangling | Pages 10-12: collection and wrangling. |
| Q6 EDA methodology | Pages 13-14: charts and SQL methods. |
| Q7 Classification methodology | Page 17: split, scaling and tuning. |
| Q8 EDA visualizations | Pages 18-23: six charts with analysis. |
| Q9 SQL results and queries | Pages 14, 24-33: ten SQL results. |
| Q10 Folium maps | Pages 15, 34-36: three map views. |
| Q11 Plotly Dash | Pages 16, 37-39: dashboard interactions. |
| Q12 Predictive results | Pages 4-6: accuracy, matrix and selection. |
| Q13 Conclusion | Page 7: findings, limits and next steps. |
| Q14 Creativity | Custom flows/charts: 10-12, 18-23. Maps: 34-39. |
| Q15 Additional insights | Page 8: three insights with numerical evidence. |

# Executive Summary

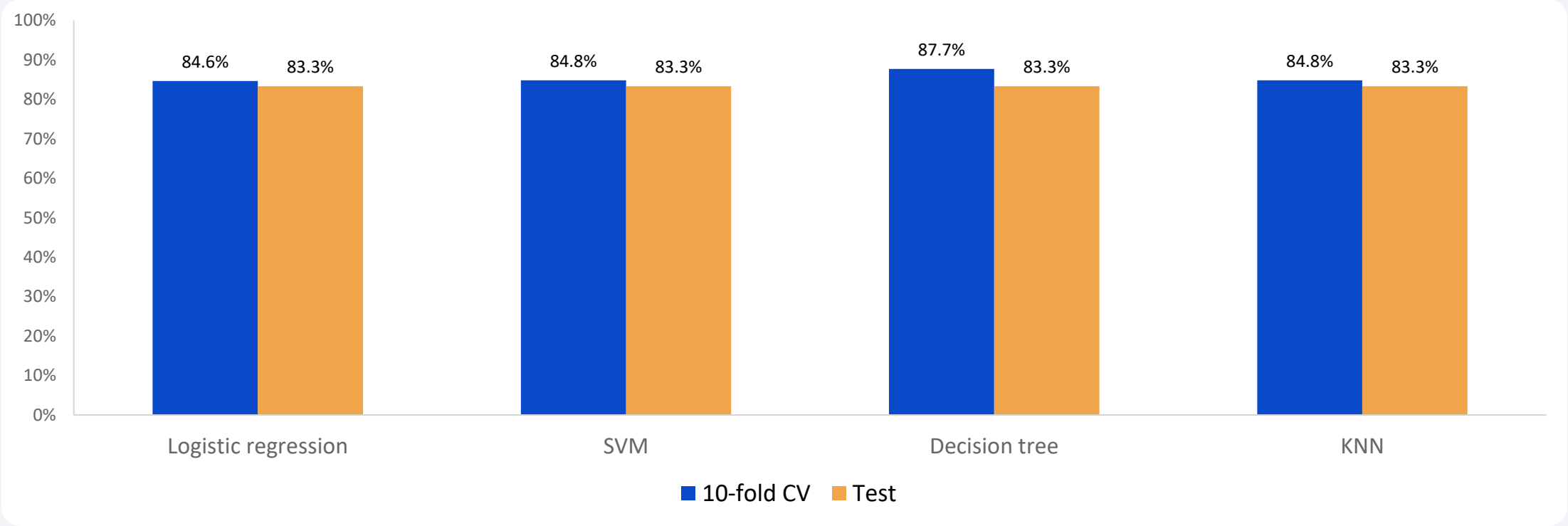
Methods: archived API data and web scraping, data cleaning, SQL and visual EDA, Folium maps, a Dash dashboard and four tuned classifiers.

Results: 60 of 90 launches have Class 1 outcomes (66.7%). KSC LC 39A has the highest observed site rate at 17/22 (77.3%).

Prediction: all four classifiers score 83.33% on 18 test launches. Decision tree leads cross-validation at 87.68%.

Interpretation: outcome definition, mission mix, small samples and preprocessing leakage limit business use of these predictions.

# Classification Accuracy




All models tie at 83.33% on the test set. The decision tree leads CV at 87.68%. The always-Class-1 test baseline is 66.67%.

# Decision Tree Confusion Matrix

| Actual / Predicted | Class 0 | Class 1 |
|---|---|---|
| Class 0 | 3 | 3 |
| Class 1 | 0 | 12 |

15/18 correct (83.33%)
Class 1 precision: 12/15 = 80%
Class 1 recall: 12/12 = 100%

3 of 6 actual Class 0 outcomes are false positives

# Model Evaluation and Selection

| Classifier | 10-fold CV accuracy | Test accuracy |
|---|---|---|
| Logistic regression | 84.64% | 83.33% |
| Support vector machine | 84.82% | 83.33% |
| Decision tree | 87.68% | 83.33% |
| K-nearest neighbors | 84.82% | 83.33% |

Decision tree is selected for its highest cross-validation accuracy: 87.68%. All four models tie at 83.33% on 18 test launches.

Split: 72 train, 18 test. Small samples and preprocessing leakage limit confidence.

https://github.com/nicholas-yap/spacex-data-science-capstone/blob/main/notebooks/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

# Conclusions

Historical results: 60/90 Class 1 outcomes (66.7%), rising to 16/19 (84.2%) in 2020.

KSC leads observed site rate at 17/22 (77.3%). CCAFS contributes the most successes, 33. Mission mix limits comparisons between sites.

All four classifiers achieve 83.33% test accuracy. Decision tree leads CV at 87.68%, but predicts success for 3 of 6 actual Class 0 outcomes.

Next steps: fit preprocessing inside cross-validation, check which features are available before launch, and test on later launches before using predictions in business decisions.

# Additional Insights

| Finding | Numerical evidence | Interpretation |
|---|---|---|
| Count differs from rate | CCAFS: 33/55 (60.0%)<br>KSC: 17/22 (77.3%) | Most successes and highest success rate occur at different sites. |
| Outcome label differs from pad landing | Class 1: 60/90 (66.7%)<br>ASDS or RTLS: 55/90 (61.1%) | Five controlled ocean outcomes add 5.6 percentage points to Class 1 rate. |
| Booster era matters | Block 5: 33/39 (84.6%)<br>Block 1: 4/19 (21.1%) | Technology and date overlap. This comparison does not isolate a causal effect. |

Block 2 has 6/6 successes, but its small sample prevents a strong reliability claim. These are historical associations, not estimates of launch price or savings.

# Introduction

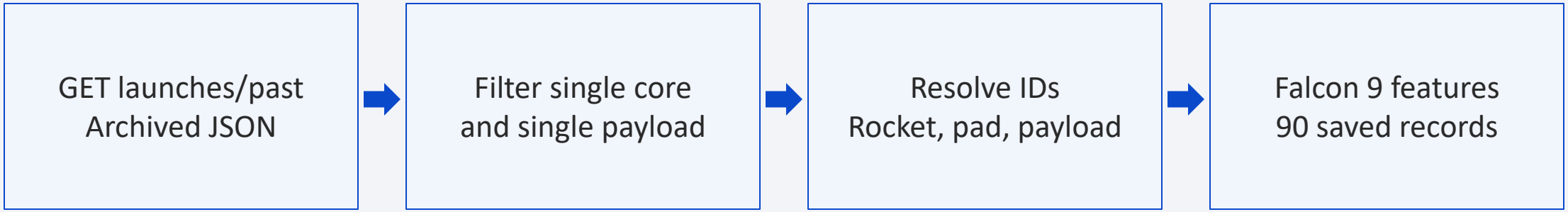
Business question: how can historical launch characteristics help estimate the chance of a successful first-stage outcome?

Reusable boosters create a reason to study landing outcomes. This project predicts the course outcome label, not launch price or cost savings.

Compare launch sites, payload, orbit and experience, then evaluate four classification models.

Scope: historical course snapshots. A successful mission and a successful first-stage outcome measure different events.

# Data Collection – SpaceX API

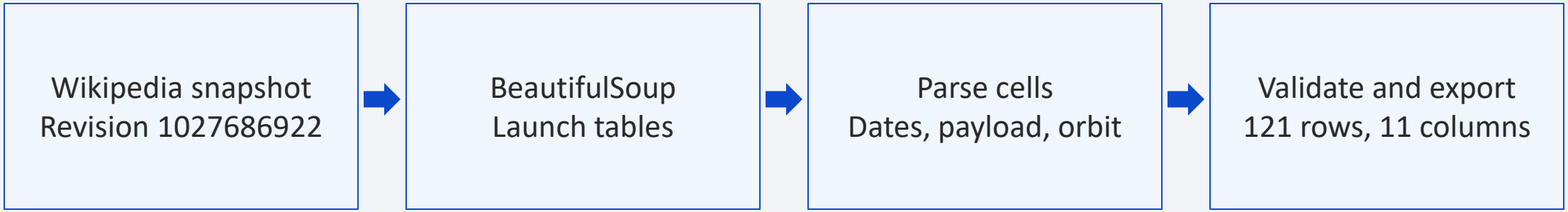



The archived launch response supports reproducible inspection. The supplied course CSV contains the enriched rocket, launchpad, payload and core fields.

The companion notebook validates the snapshot and the 90-row feature table without claiming a fresh API extraction.

https://github.com/nicholas-yap/spacex-data-science-capstone/blob/main/notebooks/01-data-collection.ipynb

# Data Collection – Web Scraping

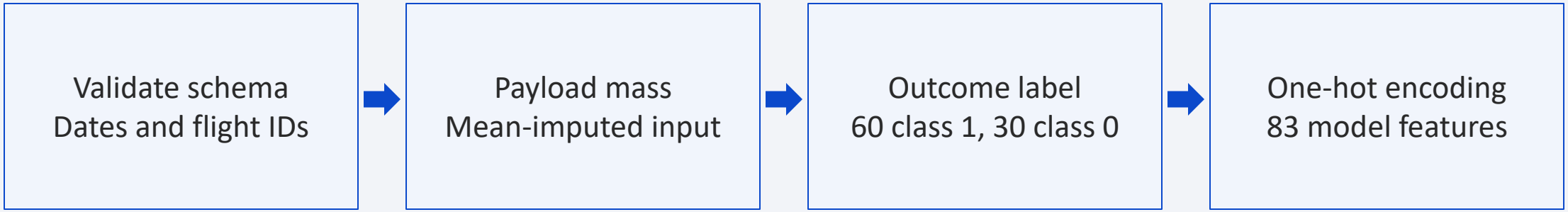



The completed notebook extracts numbered launch rows and checks flight IDs, dates and times.

The June 2021 page snapshot extends beyond the 90-row modeling sample. The scraped table remains a separate source.

https://github.com/nicholas-yap/spacex-data-science-capstone/blob/main/notebooks/jupyter-labs-webscraping.ipynb

# Data Wrangling




Class 1 includes True ASDS, True RTLS and True Ocean. Class 0 includes failed, missing and no-attempt outcomes.

Controlled ocean outcomes do not establish booster recovery. Keep this distinction when interpreting “success.”

https://github.com/nicholas-yap/spacex-data-science-capstone/blob/main/notebooks/02-wrangling-eda.ipynb

# EDA with Data Visualization

Scatter plots compare flight number and payload with launch site and orbit. Color separates the two outcome classes.

Orbit success bars show rates with launch counts so small groups remain visible.

The yearly line chart shows changes in outcome rate, with year-specific denominators.

Charts describe associations. Site, orbit, payload and technology changed together over time.

https://github.com/nicholas-yap/spacex-data-science-capstone/blob/main/notebooks/02-wrangling-eda.ipynb

# EDA with SQL

DISTINCT and LIKE identify launch-site labels and the first five CCA records.

SUM, AVG and MAX summarize payload mass and the boosters carrying the maximum.

Date and outcome filters identify the first ground landing, drone ship landings and failures in 2015.

GROUP BY and COUNT summarize mission results and rank landing outcomes for the specified period.

https://github.com/nicholas-yap/spacex-data-science-capstone/blob/main/notebooks/jupyter-labs-eda-sql-coursera_sqllite.ipynb

# Interactive Map with Folium

Site circles and labels locate the three modeled launch facilities on a continental map.

A MarkerCluster groups overlapping launches. Green marks Class 1 and red marks Class 0.

Popup details show flight, date, payload and outcome. Expanding a cluster separates markers for inspection.

Lines connect CCAFS SLC 40 to selected road, rail and coastline points. Haversine distance gives approximate straight-line separation.

https://github.com/nicholas-yap/spacex-data-science-capstone/blob/main/notebooks/03-interactive-maps.ipynb

# Dashboard with Plotly Dash

The site dropdown switches between all launches and a selected facility.

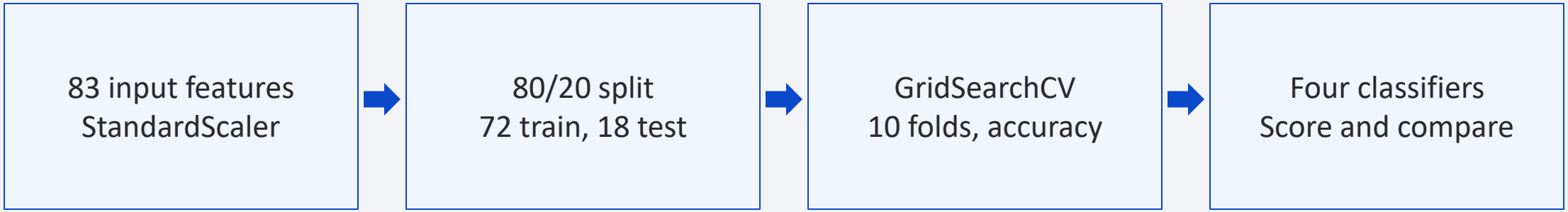
The all-sites pie chart shows each site’s contribution to successful outcomes. A selected-site pie shows its success ratio.

The payload range slider filters the outcome scatter plot. Color identifies booster block.

Hover details reveal launch date, flight number and outcome. The dashboard uses the same 90-row sample as the EDA.

https://github.com/nicholas-yap/spacex-data-science-capstone/blob/main/spacex_dashboard.py

# Predictive Analysis

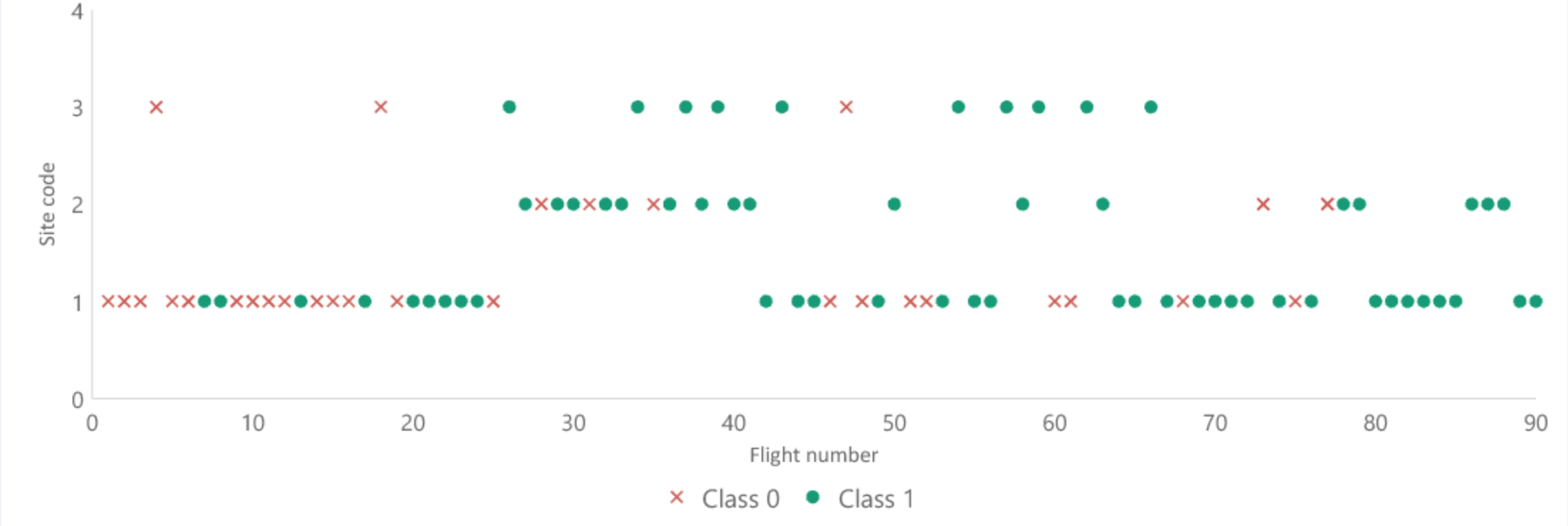



Logistic regression, SVM, decision tree and KNN use the saved lab configuration. random_state=2 fixes the train/test split.

The lab scales all 90 rows before splitting. Future evaluation should fit preprocessing inside each training fold and use a chronological holdout.

https://github.com/nicholas-yap/spacex-data-science-capstone/blob/main/notebooks/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

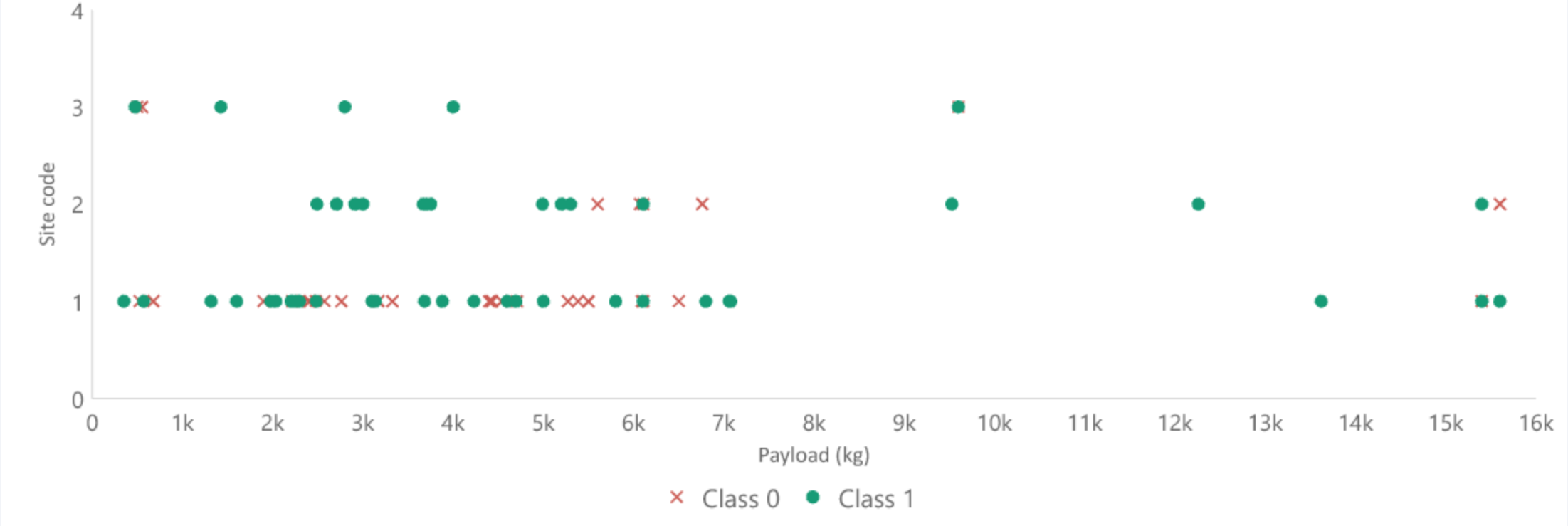
# Flight Number vs. Launch Site




1: CCAFS SLC 40 2: KSC LC 39A 3: VAFB SLC 4E

Later launches show more Class 1 outcomes. Site and flight sequence overlap, so the plot cannot isolate a site effect.

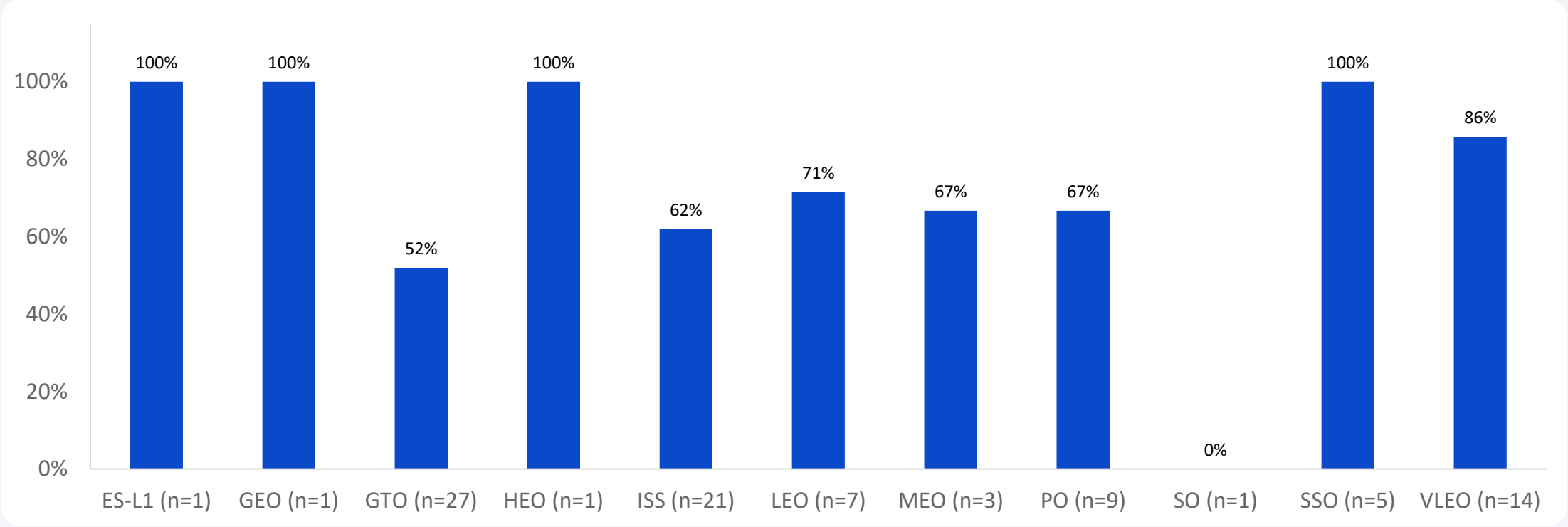
# Payload vs. Launch Site




1: CCAFS SLC 40 2: KSC LC 39A 3: VAFB SLC 4E

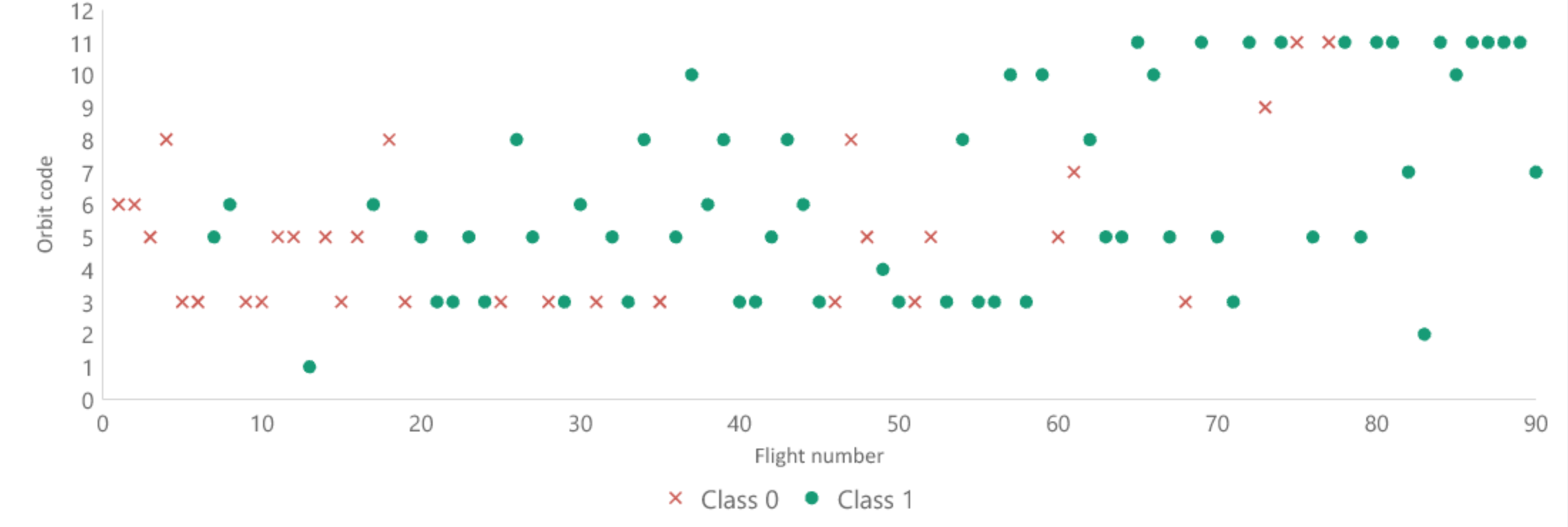
CCAFS SLC 40 covers the broadest history. Both outcome classes occur across payload ranges, and some payload values are mean-imputed.

# Success Rate vs. Orbit Type




SSO records 5/5 successes. GEO, HEO and ES-L1 each have only one launch, so their 100% rates provide little evidence.

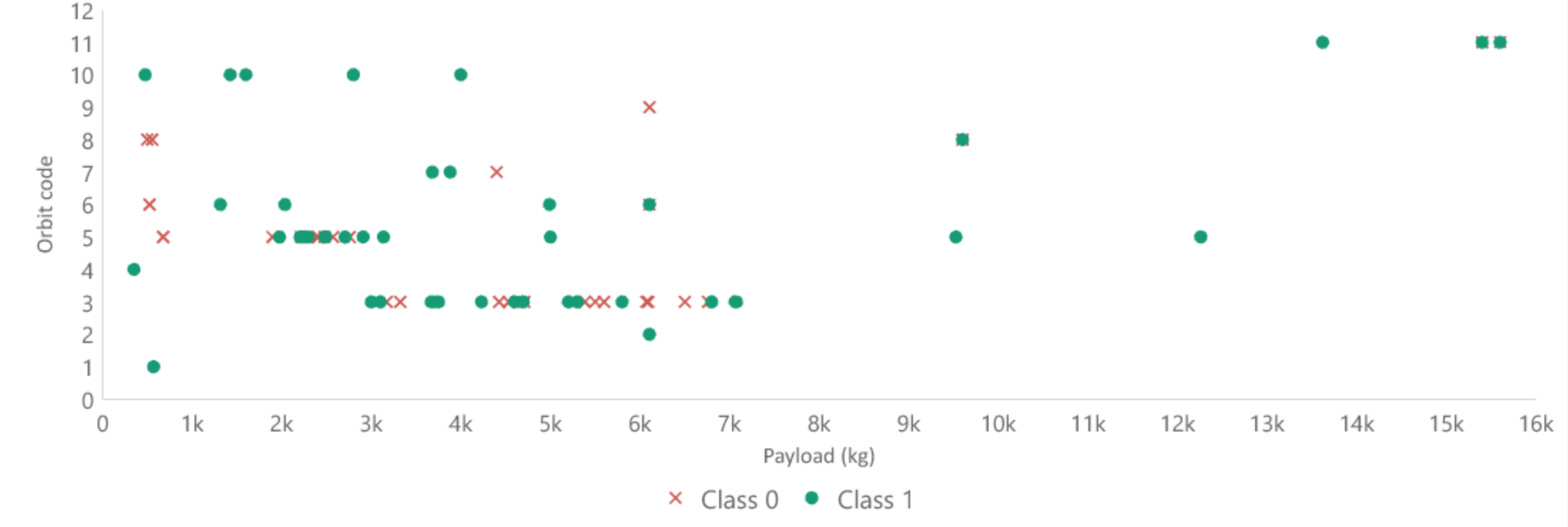
# Flight Number vs. Orbit Type




1: ES-L1 2: GEO 3: GTO 4: HEO 5: ISS 6: LEO 7: MEO 8: PO 9: SO 10: SSO 11: VLEO

VLEO launches cluster late in the sample. Differences by orbit may partly reflect changes in booster technology and experience.

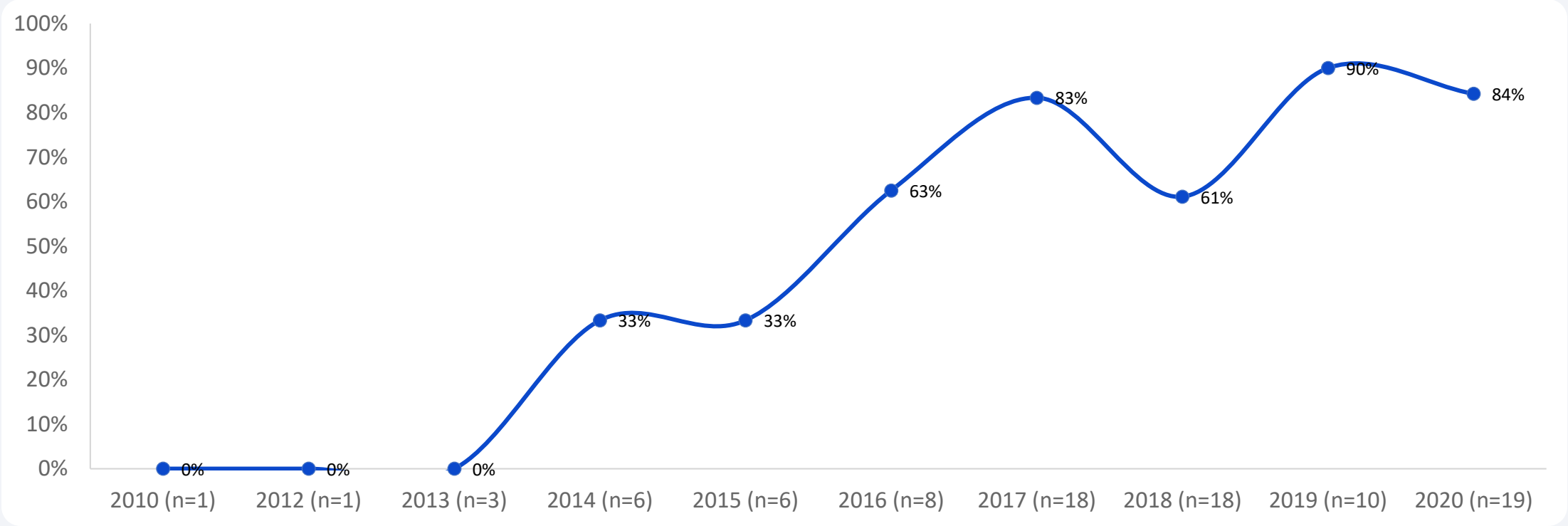
# Payload vs. Orbit Type




1: ES-L1 2: GEO 3: GTO 4: HEO 5: ISS 6: LEO 7: MEO 8: PO 9: SO 10: SSO 11: VLEO

GTO has 14/27 Class 1 outcomes (51.9%). VLEO has 12/14 (85.7%). Payload alone does not explain the difference.

# Yearly Outcome Trend




The rate rises from 0% in early years to 84.2% in 2020, with a dip in 2018. The sample contains no 2011 launch.

# All Launch Site Names

| Launch site |
| --- |
| CCAFS LC-40 |
| CCAFS SLC-40 |
| KSC LC-39A |
| VAFB SLC-4E |

The SQL source contains four labels. CCAFS LC-40 and CCAFS SLC-40 appear separately in this snapshot.

# Launch Sites Beginning with ‘CCA’

| Date | Booster version | Payload (kg) | Site |
|---|---|---|---|
| 2010-06-04 | F9 v1.0  B0003 | 0 | CCAFS LC-40 |
| 2010-12-08 | F9 v1.0  B0004 | 0 | CCAFS LC-40 |
| 2012-05-22 | F9 v1.0  B0005 | 525 | CCAFS LC-40 |
| 2012-10-08 | F9 v1.0  B0006 | 500 | CCAFS LC-40 |
| 2013-03-01 | F9 v1.0  B0007 | 677 | CCAFS LC-40 |

The earliest five CCA records span June 2010 to March 2013. The query orders by date and UTC time before LIMIT 5.

# Total NASA (CRS) Payload Mass

## 45,596 kg

SUM(PAYLOAD_MASS__KG_)
WHERE Customer = 'NASA (CRS)'

This is the NASA Commercial Resupply Services subset. It does not include every customer label containing NASA.

# Average Payload Mass for F9 v1.1

## 2,928.4 kg

```
AVG(PAYLOAD_MASS__KG_)
WHERE Booster_Version = 'F9 v1.1'
```

The exact version label matches five rows. Including every label beginning with F9 v1.1 would answer a different query.

# First Successful Ground Landing

## 22 December 2015

MIN(Date)
WHERE TRIM(Landing_Outcome) = 'Success (ground pad)'

This is the earliest successful ground-pad landing recorded in the SQL snapshot.

# Drone Ship Landings at 4,000–6,000 kg

| Booster version |
| --- |
| F9 FT  B1021.2 |
| F9 FT  B1031.2 |
| F9 FT B1022 |
| F9 FT B1026 |

Four boosters meet both conditions: successful drone ship landing and payload strictly above 4,000 kg and below 6,000 kg.

# Mission Outcomes

| Mission result | Missions |
|---|---|
| Success | 100 |
| Failure | 1 |

100/101 missions count as successful. “Success (payload status unclear)” is included. Mission results differ from landing outcomes.

# Boosters Carrying the Maximum Payload

| Booster | Payload (kg) | Booster | Payload (kg) |
|---|---|---|---|
| F9 B5 B1048.4 | 15600 | F9 B5 B1051.4 | 15600 |
| F9 B5 B1048.5 | 15600 | F9 B5 B1051.6 | 15600 |
| F9 B5 B1049.4 | 15600 | F9 B5 B1056.4 | 15600 |
| F9 B5 B1049.5 | 15600 | F9 B5 B1058.3 | 15600 |
| F9 B5 B1049.7 | 15600 | F9 B5 B1060.2 | 15600 |
| F9 B5 B1051.3 | 15600 | F9 B5 B1060.3 | 15600 |

Twelve booster-flight labels share the maximum payload of 15,600 kg in this snapshot.

# Drone Ship Landing Failures in 2015

| Month | Landing outcome | Booster | Site |
|---|---|---|---|
| January | Failure (drone ship) | F9 v1.1 B1012 | CCAFS LC-40 |
| April | Failure (drone ship) | F9 v1.1 B1015 | CCAFS LC-40 |

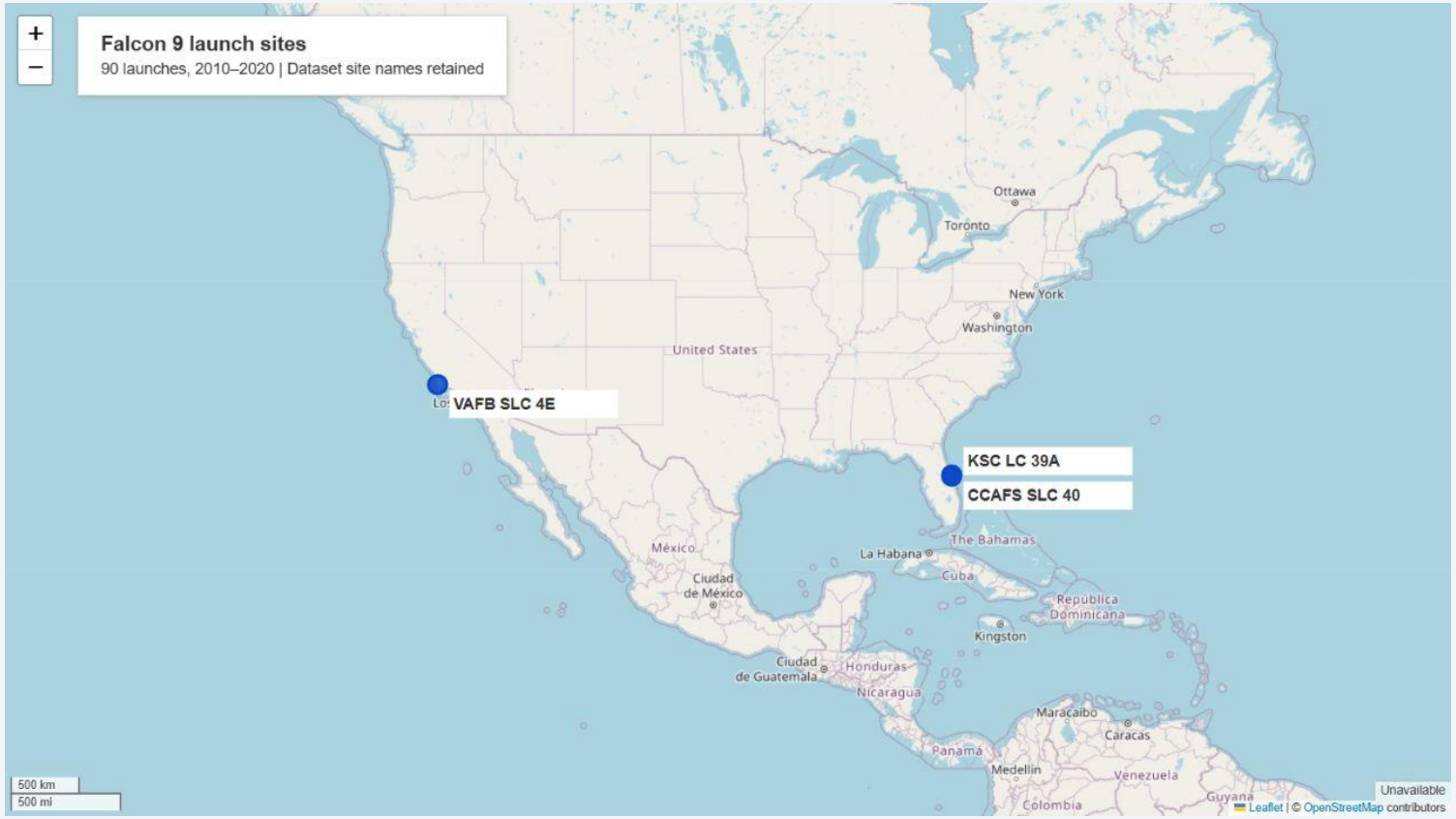
Two 2015 records show failed drone ship landings: January (B1012) and April (B1015), both at CCAFS LC-40.

# Landing Outcome Ranking, 2010–2017

| Landing outcome | Count |
|---|---|
| No attempt | 10 |
| Failure (drone ship) | 5 |
| Success (drone ship) | 5 |
| Controlled (ocean) | 3 |
| Success (ground pad) | 3 |
| Failure (parachute) | 2 |
| Uncontrolled (ocean) | 2 |
| Precluded (drone ship) | 1 |

Inclusive period: 4 June 2010 to 20 March 2017. No attempt is most frequent (10). Drone ship successes and failures tie at five.

# Launch Sites on the US Coasts

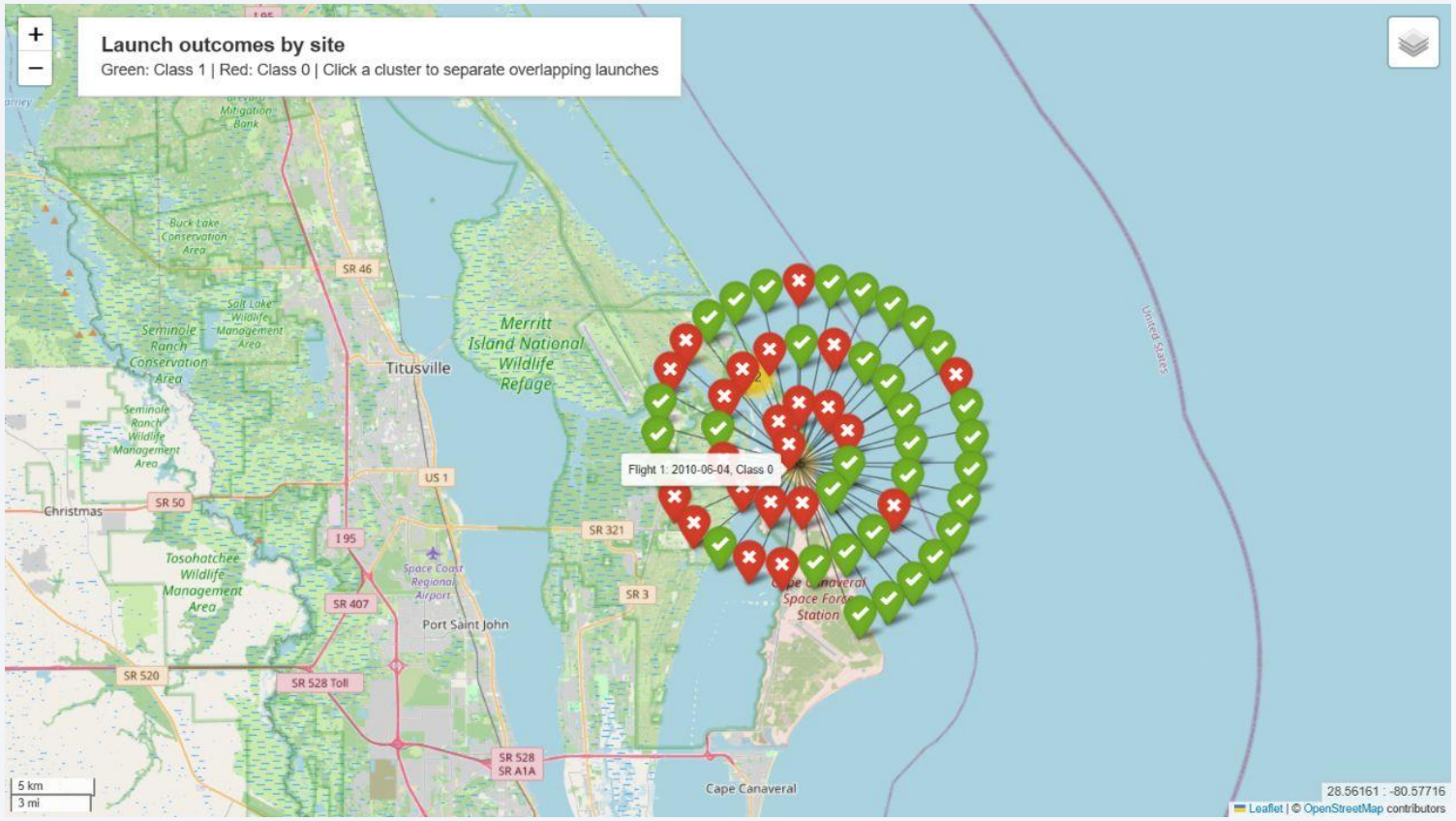



Florida
CCAFS SLC 40
KSC LC 39A

California
VAFB SLC 4E

All three facilities are coastal. The two Florida pads lie close together. Location alone does not explain landing performance.

# Launch Outcomes at CCAFS SLC 40



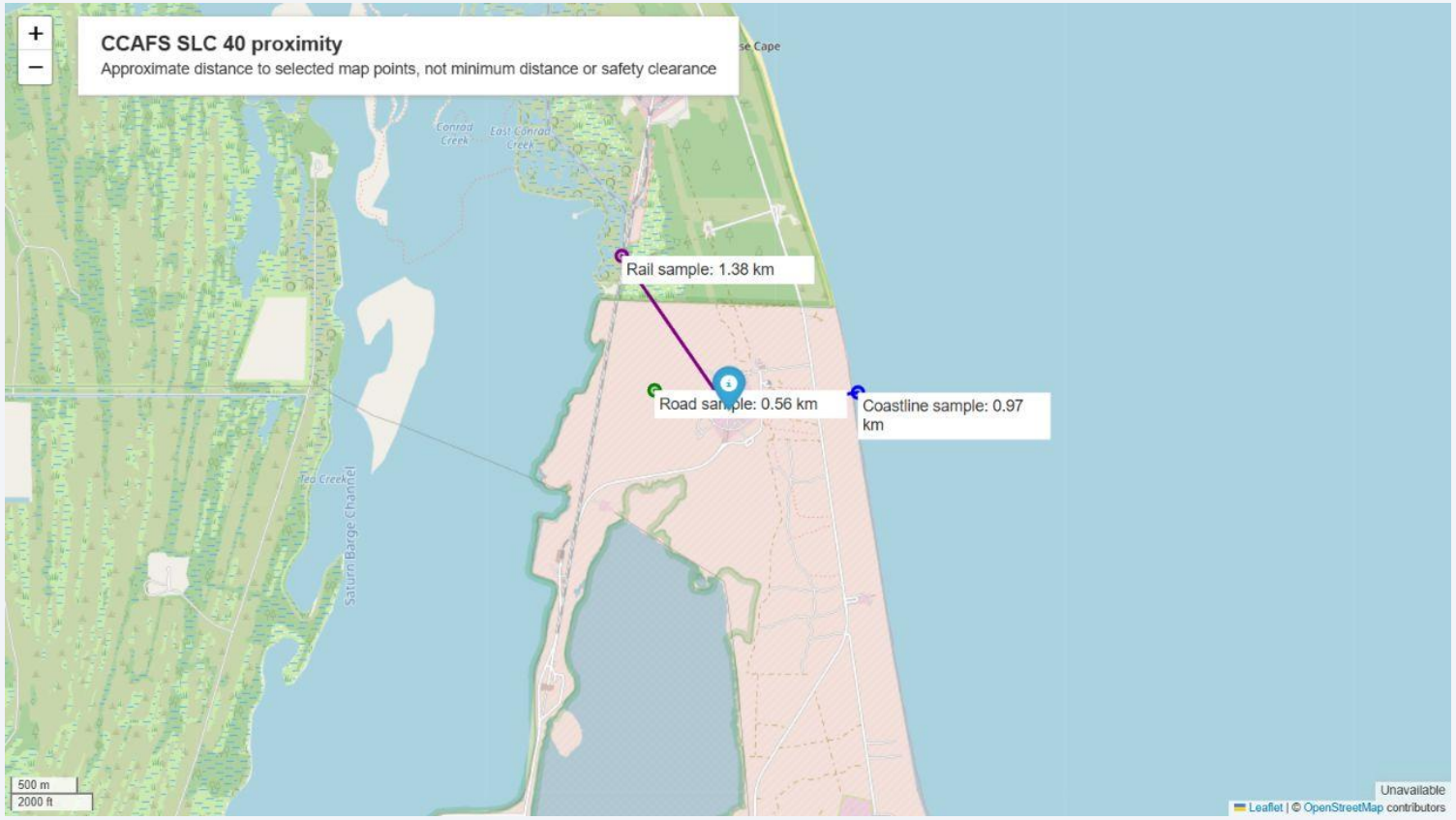

55 launches

33 Class 1
22 Class 0

60.0% success

Expanded markers separate overlapping records around the same launch site. Their displayed offsets are not actual launch locations.

# CCAFS SLC 40 and Nearby Features

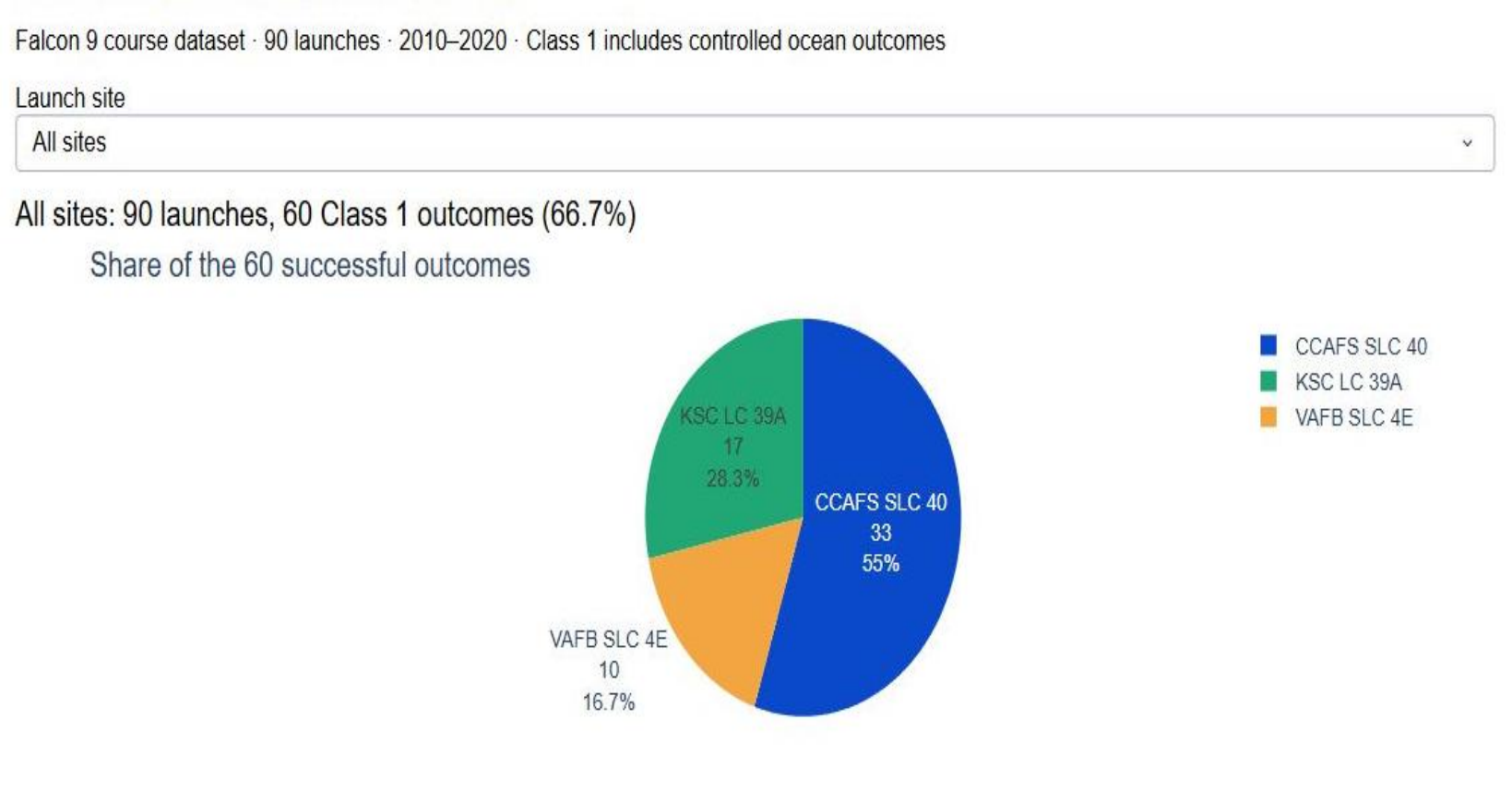



Selected points

Coast: 0.97 km
Road: 0.56 km
Rail: 1.38 km

Approximate straight-line distances to selected map points. These are not surveyed minimum distances or safety clearances.

# Dashboard: Successful Outcomes by Site



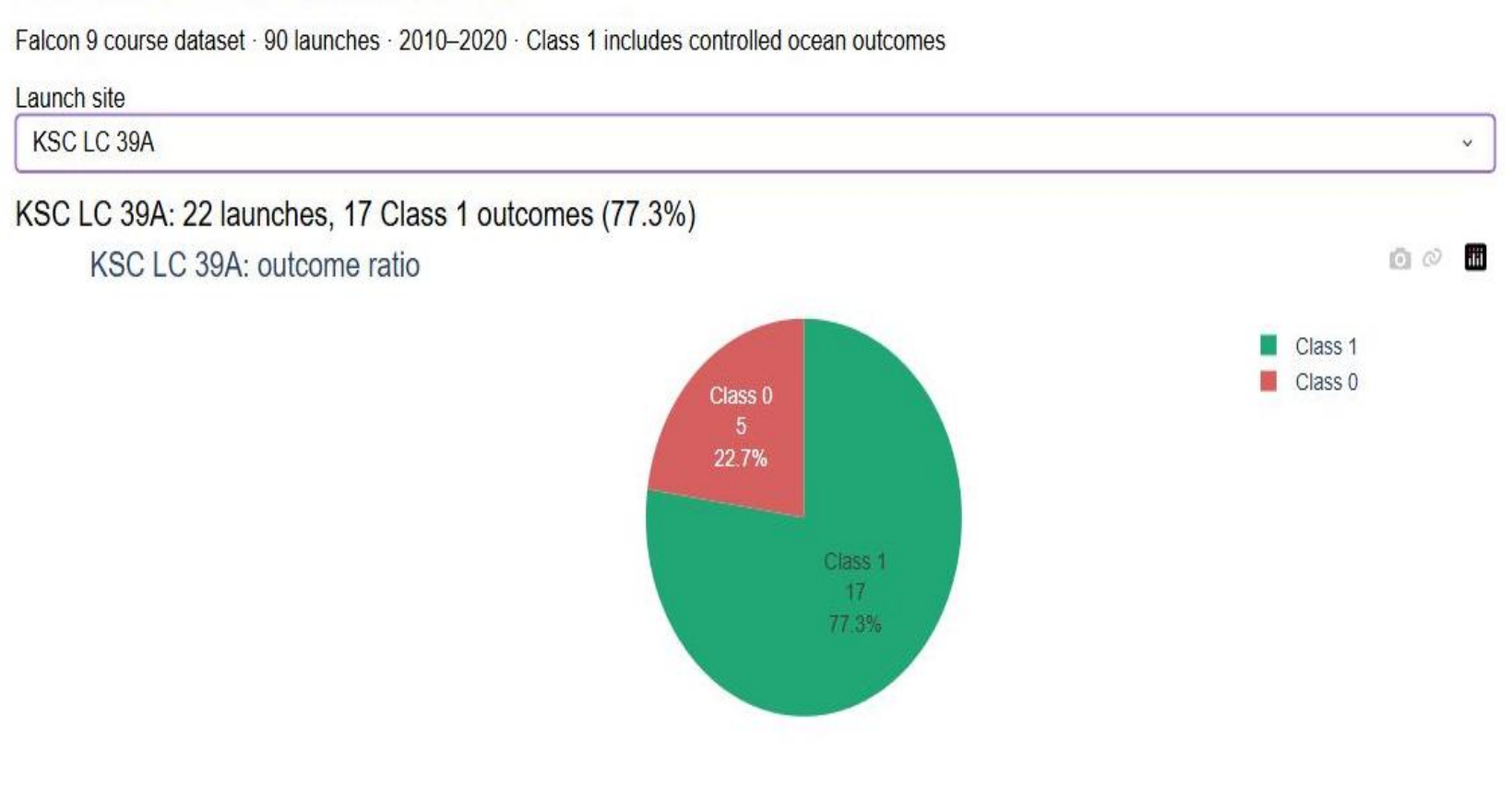

60 successes

CCAFS: 33
KSC: 17
VAFB: 10

CCAFS contributes 55% of successes.

The pie compares successful-outcome counts. A larger slice does not imply a higher success rate.

# Dashboard: Highest Observed Site Rate



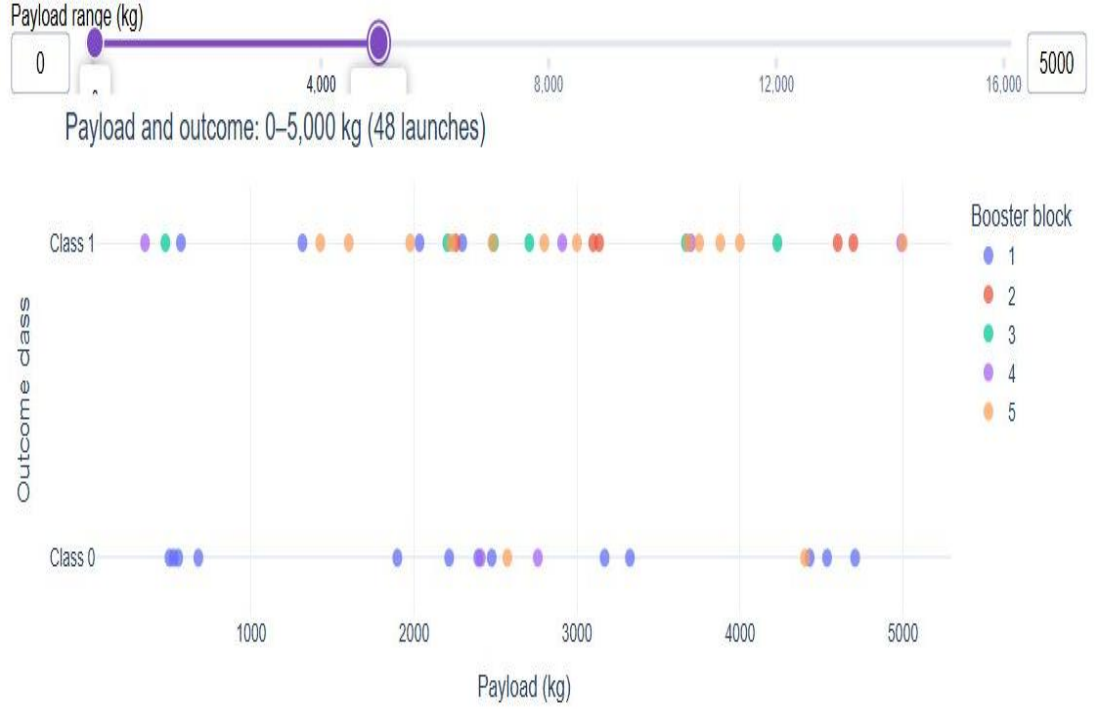

KSC LC 39A

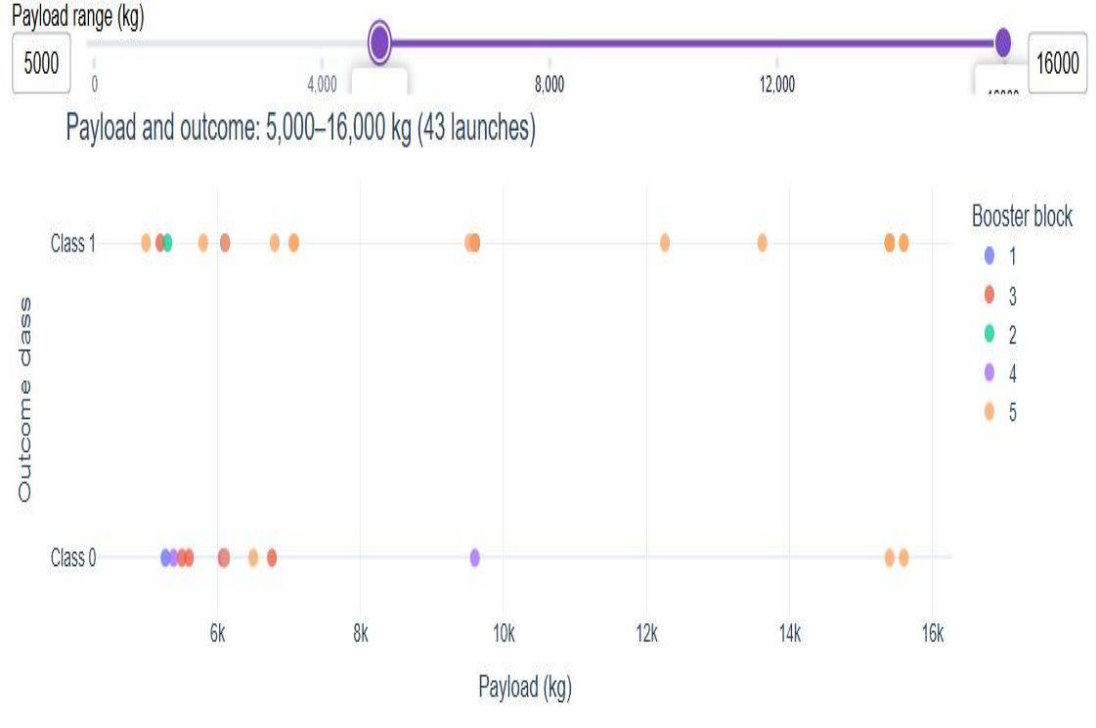
17 successes
5 class 0

77.3% success
22 launches

KSC narrowly exceeds VAFB's 76.9% (10/13). These small samples do not establish a reliable ranking of site quality.

# Dashboard: Payload Range Filters




Two slider selections isolate different payload bands. Booster block and flight date also vary, so a payload association is not a causal effect.

# Appendix and Reproducible Sources

## Public repository

https://github.com/nicholas-yap/spacex-data-science-capstone

| | |
|---|---|
| notebooks/ | Collection, scraping, wrangling, EDA, SQL, maps and classification |
| data/ | Historical course snapshots and scraped records |
| maps/ | Interactive HTML maps |
| spacex_dashboard.py | Runnable dashboard on port 8051 |